

Lab10

Niam Shasha A17805610

Background

The main repository of high-resolution structural data on biomolecules is called the **Protein Data Bank**

PDB Statistics

Read a CSV file from a current PDB stats obtained from

```
pdb <- read.csv("Data Export Summary.csv")
pdb
```

	Molecular.Type	X.ray	EM	NMR	Integrative	Multiple.methods
1	Protein (only)	180,758	23,111	12,813	348	229
2	Protein/Oligosaccharide	10,488	3,741	34	8	11
3	Protein/NA	9,205	6,751	287	26	8
4	Nucleic acid (only)	3,154	250	1,578	3	15
5	Other	178	27	35	4	0
6	Oligosaccharide (only)	11	0	6	0	1
	Neutron	Other	Total			
1	84	32	217,375			
2	1	0	14,283			
3	0	0	16,277			
4	3	1	5,004			
5	0	0	244			
6	0	4	22			

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy

Around 80.49% of PDB structures are solved by X-Ray and 13.38% by electron microscopy

```
pdb$X.ray
```

```
[1] "180,758" "10,488" "9,205" "3,154" "178" "11"
```

The Print out above is `pdb$X.ray` is “character” not “numeric”. Therefore, I can’t do math w/ it. We need to fix this...

```
pdb$Neutron
```

```
[1] 84 1 0 3 0 0
```

```
#We want to remove commas:  
x <- pdb$X.ray  
tmp <- sub(",", "", x)  
sum(as.numeric(tmp))
```

```
[1] 203794
```

We could make a function to do this:

```
rm.comma <- function (x){  
  tmp <- sub(",", "", x)  
  sum(as.numeric(tmp))  
}
```

```
n.tot <- rm.comma(pdb$Total)  
n.xray <- rm.comma(pdb$X.ray)  
n.em <- rm.comma(pdb$EM)
```

```
n.xray/n.tot * 100
```

```
[1] 80.48577
```

```
n.em/n.tot * 100
```

```
[1] 13.38046
```

Different import function for this cvs file thhat speaks American (i.e deals w/ commas in umbers in a comma seperated value file).

```
library(readr)

pdb <- read_csv("Data Export Summary.csv")
```

```
Rows: 6 Columns: 9
-- Column specification -----
Delimiter: ","
chr (1): Molecular Type
dbl (4): Integrative, Multiple methods, Neutron, Other
num (4): X-ray, EM, NMR, Total

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
pdb$Total
```

```
[1] 217375 14283 16277 5004 244 22
```

```
n.tot <- (pdb$Total)
n.xray <- sum(pdb$`X-ray`)
sum(pdb$`X-ray`) / n.tot
```

```
[1] 0.9375227 14.2682910 12.5203662 40.7262190 835.2213115
[6] 9263.3636364
```

```
n.xray / n.tot * 100
```

```
[1] 93.75227 1426.82910 1252.03662 4072.62190 83522.13115
[6] 926336.36364
```

```
n.em / n.tot * 100
```

```
[1] 15.58597 237.20507 208.14646 677.05835 13885.24590
[6] 154000.00000
```

```
pdb$Total[1]
```

```
[1] 217375
```

```
{  
tmp <- sub(",", "", x)  
sum(as.numeric(tmp))  
}
```

```
[1] 203794
```

The total number of protein sequences is 202,556,314

```
sum(pdb$Total[1:3]) / sum(pdb$Total) * 100
```

```
[1] 97.91868
```

```
protein <- 217375 + 14283 + 16277  
total <- 217375 + 14283 + 16277 + 5004 + 244 + 22  
  
protein / total * 100
```

```
[1] 97.91868
```

Q2: What proportion of structures in the PDB are protein?

Around 98% in the PDB are protein

```
217375/2025556314 * 100
```

```
[1] 0.01073162
```

Key-Point: We have a very small structural coverage of known proteins (-0.1%).
Most structures we know about (80%) come from one method.

Visualizing PDB data w/ Mol-star

Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

More than 4000 structures

Main stand alone web version w/ all features is at <https://molstar.org/viewer/>



Figure 1: The HIV-protease enzyme is a homodimer w/ two chains

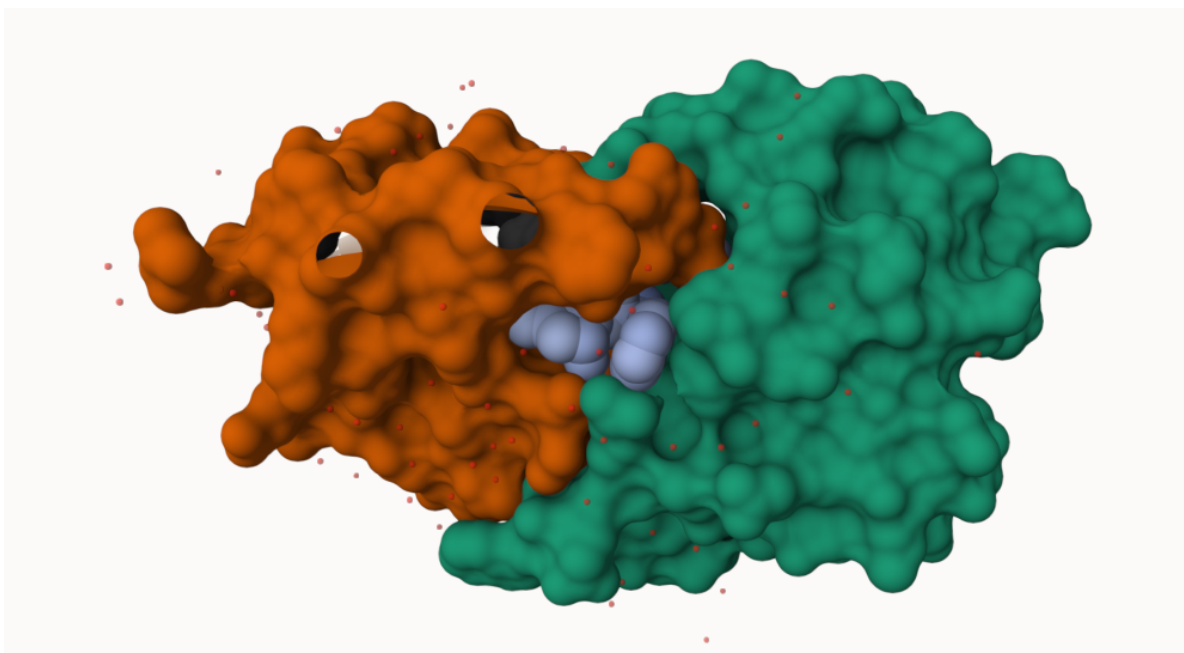


Figure 2: Surface display showing the binding cleft site for the inhibitor

Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

This is because only the Oxygen atom can be detected as Hydrogen atoms are too small for the X-ray crystallography at this resolution to detect

Q6: Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend “Ball & Stick” for these side-chains). Add this figure to your Quarto document.

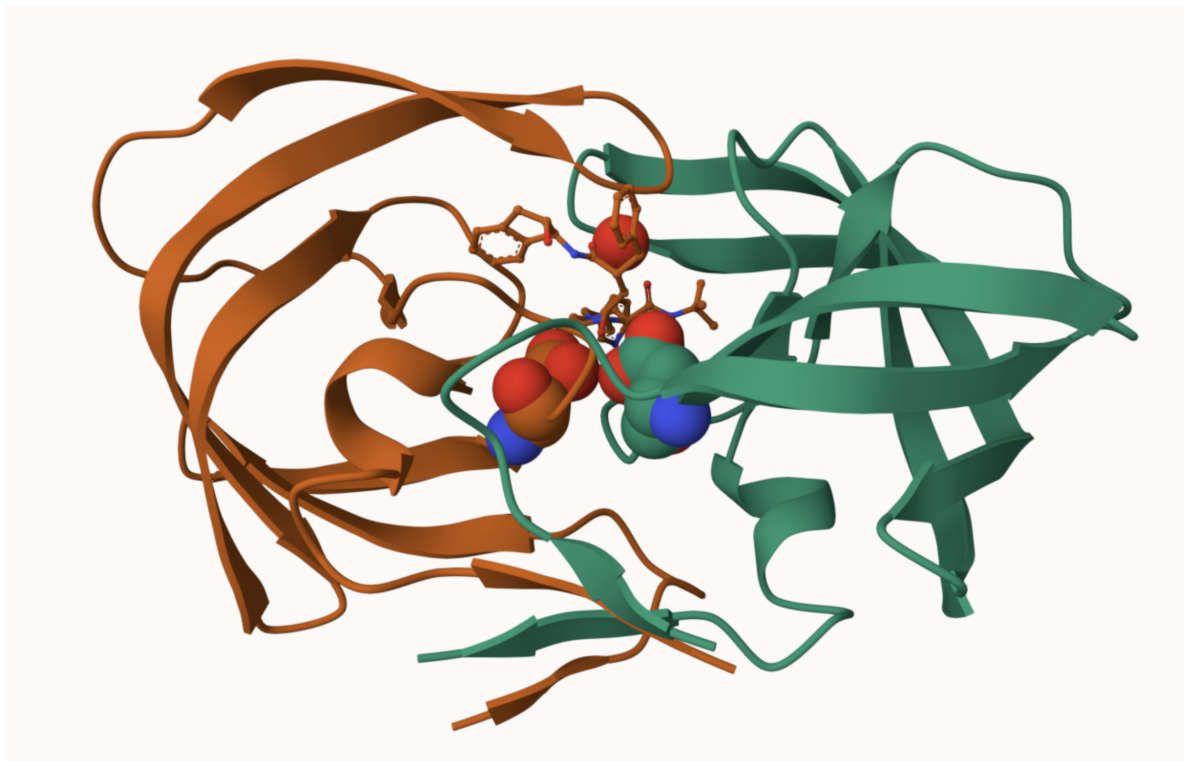


Figure 3: catalytic ASP25 amino acids and water molecule binding site

Getting started with the Bio3D package

```
library(bio3d)
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
pdb
```

```
Call: read.pdb(file = "1hsg")
```

```
Total Models#: 1
```

```
Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)
```



```
Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
```

```
Non-protein/nucleic Atoms#: 172 (residues: 128)
Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]
```

Protein sequence:

```
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF
```

```
+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call
```

```
attributes(pdb)
```

```
$names
```

```
[1] "atom" "xyz" "seqres" "helix" "sheet" "calpha" "remark" "call"
```

```
$class
```

```
[1] "pdb" "sse"
```

```
head(pdb$atom, 5)
```

	type	eleno	elety	alt	resid	chain	resno	insert	x	y	z	o	b
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	29.361	39.686	5.862	1	38.10
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	30.307	38.663	5.319	1	40.62
3	ATOM	3	C	<NA>	PRO	A	1	<NA>	29.760	38.071	4.022	1	42.64
4	ATOM	4	O	<NA>	PRO	A	1	<NA>	28.600	38.302	3.676	1	43.40
5	ATOM	5	CB	<NA>	PRO	A	1	<NA>	30.508	37.541	6.342	1	37.87

	segid	elesy	charge
1	<NA>	N	<NA>
2	<NA>	C	<NA>
3	<NA>	C	<NA>
4	<NA>	O	<NA>
5	<NA>	C	<NA>

There are lots of functions that can work with these `pdb` objects

```
head(pdbseq(pdb))
```

```
  1    2    3    4    5    6  
"P" "Q" "I" "T" "L" "W"
```

We can have a quick interactive view of any of these 'pdb' objects:

```
library(bio3dview)
```

```
#view.pdb(pdb)
```

custom view

```
#view.pdb(pdb, colorScheme="see", backgroundColor ="black")
```

Q. Create a custom view highlighting the active site ASP ('resno=25'), the two chains (in your choice of colors) and the ligand all on a custom color background?

```
library(NGLViewer)
```

```
active.site <- atom.select(pdb,resno=25)
```

```
#view.pdb(pdb,  
#       cols = c("red", "blue"),  
#       highlight = active.site,  
#       highlight.style = "spacefill",  
#       backgroundColor= "pink") |>  
#setRock()
```

Predict the Flexibility of a given structure

Let's do a Normal Mode Analysis (NMA) to predict the flexibility of a given pdb object:

```
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file

PDB has ALT records, taking A only, rm.alt=TRUE

A quick structure summary

```
adk
```

```
Call: read.pdb(file = "6s36")
```

```
Total Models#: 1
```

```
Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)
```

```
Protein Atoms#: 1654 (residues/Calpha atoms#: 214)
```

```
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
```

```
Non-protein/nucleic Atoms#: 244 (residues: 244)
```

```
Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]
```

```
Protein sequence:
```

```
MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLV  
DELVIALVKERIAQEDCRNGFLLDGFPRTPQADAMKEAGINVDYVLEFDVPDELIVDKI  
VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG  
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
```

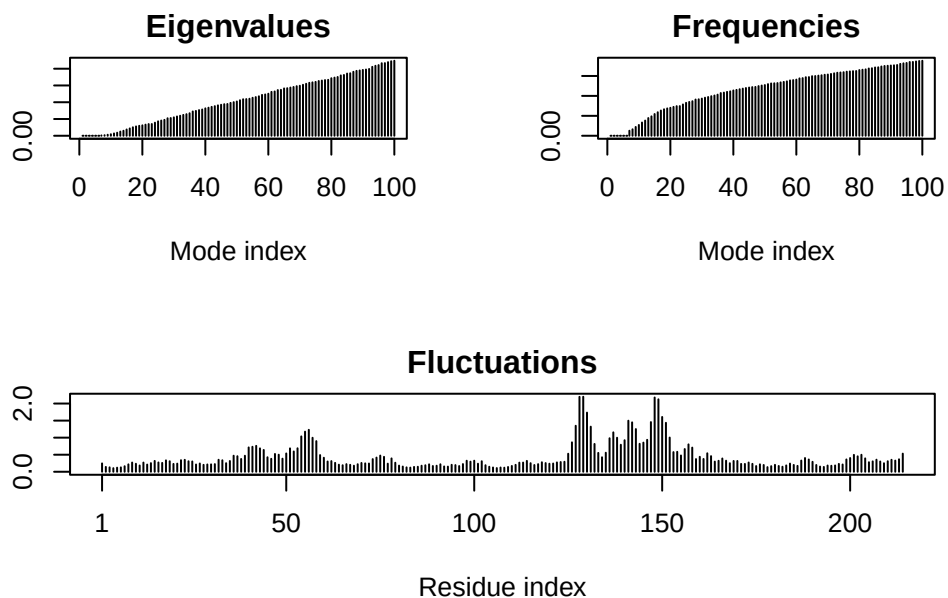
```
+ attr: atom, xyz, seqres, helix, sheet,  
      calpha, remark, call
```

```
m <- nma(adk)
```

```
Building Hessian... Done in 0.009 seconds.
```

```
Diagonalizing Hessian... Done in 0.18 seconds.
```

```
plot(m)
```



View the results

```
#view.nma(m)
```

Write out the results in Mol-star:

```
mktrj(m, file="nma.pdb")
```

Comparative analysis of the ADK family

Q10. Which of the packages above is found only on BioConductor and not CRAN?

msa is the package only on Bioconductor >Q11. Which of the above packages is not found on BioConductor or CRAN?:

bio3d#view not found on BioConductor or CRAN >Q12. True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket?

True

Our first step is find a sequence for this family using database ID "1ake_A"

```
id <- "1ake_A"
aa <- get.seq(id)
```

Warning in get.seq(id): Removing existing file: seqs.fasta

Fetching... Please wait. Done.

```
aa
```

```

      1      .      .      .      .      .      .      60
pdb|1AKE|A  MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAAVKSGSELGKQAKDIMDAGKLV
      1      .      .      .      .      .      .      60

      61      .      .      .      .      .      .      120
pdb|1AKE|A  DELVIALVKERIAQEDCRNGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVDRI
      61      .      .      .      .      .      .      120

     121      .      .      .      .      .      .      180
pdb|1AKE|A  VGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTRKDDQEETVRKRLVEYHQMTAPLIG
     121      .      .      .      .      .      .      180

     181      .      .      .      214
pdb|1AKE|A  YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
     181      .      .      .      214
```

Call:

```
read.fasta(file = outfile)
```

Class:

```
fasta
```

Alignment dimensions:

```
1 sequence rows; 214 position columns (214 non-gap, 0 gap)
```

```
+ attr: id, ali, call
```

Q13. How many amino acids are in this sequence, i.e. how long is this sequence?

there are 214 amino acids

```
#blast <- blast.pdb(aa)
```

```
#head(blast$hit.tbl)
```

```
#hits <- plot(blast)
```

```
hits <- NULL
```

```
hits$ pdb.id <-
```

```
hits$ pdb.id <- c('1AKE_A','6S36_A','6RZE_A','3HPR_A','1E4V_A','5EJE_A','1E4Y_A','3X2S_A','6HAM_A','8PVW_A','4K46_A','4NP6_A')
```

```
hits$ pdb.id
```

```
[1] "1AKE_A" "6S36_A" "6RZE_A" "3HPR_A" "1E4V_A" "5EJE_A" "1E4Y_A" "3X2S_A"  
[9] "6HAM_A" "8PVW_A" "4K46_A" "4NP6_A"
```

```
files <- get.pdb(hits$ pdb.id, path="pdbs", split=TRUE, gzip=TRUE)
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/1AKE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/6S36.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/6RZE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/3HPR.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/1E4V.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/5EJE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/1E4Y.pdb.gz exists. Skipping download
```

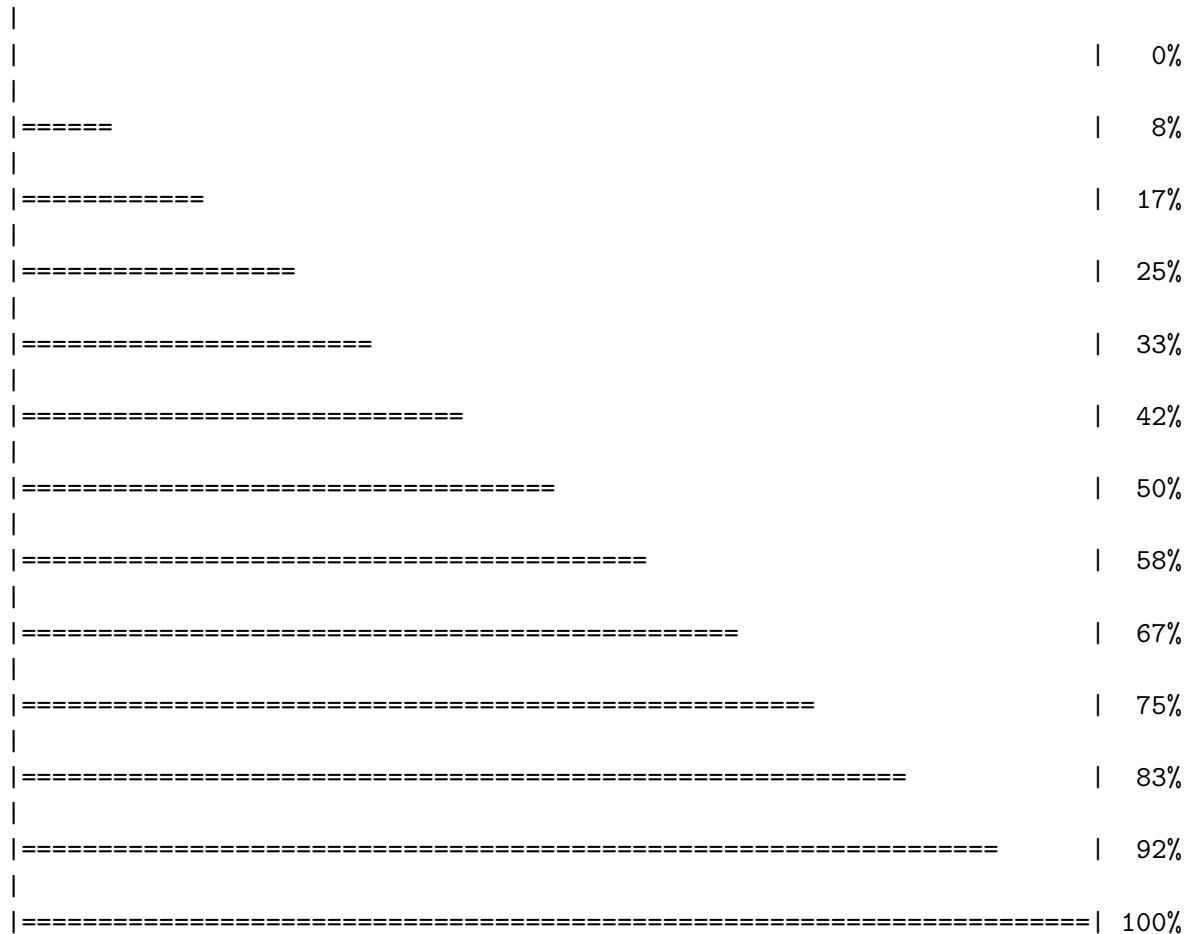
Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3X2S.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAM.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8PVW.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4K46.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4NP6.pdb.gz exists. Skipping download



Align and superpose all the ADK structures

```
pdb<- pdbaln(files, fit = TRUE, exefile="msa")
```

Reading PDB files:

```
pdb/split_chain/1AKE_A.pdb
pdb/split_chain/6S36_A.pdb
pdb/split_chain/6RZE_A.pdb
pdb/split_chain/3HPR_A.pdb
pdb/split_chain/1E4V_A.pdb
pdb/split_chain/5EJE_A.pdb
pdb/split_chain/1E4Y_A.pdb
pdb/split_chain/3X2S_A.pdb
pdb/split_chain/6HAM_A.pdb
pdb/split_chain/8PVW_A.pdb
pdb/split_chain/4K46_A.pdb
pdb/split_chain/4NP6_A.pdb
```

```
    PDB has ALT records, taking A only, rm.alt=TRUE
.    PDB has ALT records, taking A only, rm.alt=TRUE
.    PDB has ALT records, taking A only, rm.alt=TRUE
.    PDB has ALT records, taking A only, rm.alt=TRUE
..   PDB has ALT records, taking A only, rm.alt=TRUE
...   PDB has ALT records, taking A only, rm.alt=TRUE
.    PDB has ALT records, taking A only, rm.alt=TRUE
.    PDB has ALT records, taking A only, rm.alt=TRUE
..
```

Extracting sequences

```
pdb/seq: 1    name: pdb/split_chain/1AKE_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2    name: pdb/split_chain/6S36_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3    name: pdb/split_chain/6RZE_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 4    name: pdb/split_chain/3HPR_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5    name: pdb/split_chain/1E4V_A.pdb
pdb/seq: 6    name: pdb/split_chain/5EJE_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7    name: pdb/split_chain/1E4Y_A.pdb
pdb/seq: 8    name: pdb/split_chain/3X2S_A.pdb
```



```

pdb/seq: 9   name: pdbs/split_chain/6HAM_A.pdb
      PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 10  name: pdbs/split_chain/8PVW_A.pdb
      PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 11  name: pdbs/split_chain/4K46_A.pdb
      PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 12  name: pdbs/split_chain/4NP6_A.pdb

```

pdbs

```

[Truncated_Name:1] 1AKE_A.pdb      1      .      .      .      40
--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:2] 6S36_A.pdb      1      .      .      .      40
--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:3] 6RZE_A.pdb      1      .      .      .      40
--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:4] 3HPR_A.pdb      1      .      .      .      40
--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:5] 1E4V_A.pdb      1      .      .      .      40
--MRIILLGAPVAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:6] 5EJE_A.pdb      1      .      .      .      40
--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:7] 1E4Y_A.pdb      1      .      .      .      40
--MRIILLGALVAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:8] 3X2S_A.pdb      1      .      .      .      40
--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:9] 6HAM_A.pdb      1      .      .      .      40
--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:10] 8PVW_A.pdb      1      .      .      .      40
--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:11] 4K46_A.pdb      1      .      .      .      40
--MRIILLGAPGAGKGTQAQFIMAKFGIPQISTGDMLRAA
[Truncated_Name:12] 4NP6_A.pdb      1      .      .      .      40
NAMRIILLGAPGAGKGTQAQFIMEKFGIPQISTGDMLRAA
*****  ***** *~*****
1      .      .      .      40

41      .      .      .      80
VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
VKSGSELGKQAKDIMDACKLVDELVIALVKERIAQEDCR
VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
VKSGSELGKQAKDIMDCGKLVDELVIALVKERIAQEDSR
IKSGSELGKQAKDIMDAGKLVDEIIIALVKERICQEDSR
VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
IKAGTELGKQAKSVIDAGQLVSDDIILGLVKERIAQDDCA
IKAGTELGKQAKAVIDAGQLVSDDIILGLIKERIAQADCE
~* *~***** ^~*  **~*~^~^~*~**** * *
41      .      .      .      80

```

	81	.	.	.	120
[Truncated_Name:1] 1AKE_A.pdb	NGFLLDGFPR	TIPQADAMKE	AGINVDYVLE	FDVPDELIVD	
[Truncated_Name:2] 6S36_A.pdb	NGFLLDGFPR	TIPQADAMKE	AGINVDYVLE	FDVPDELIVD	
[Truncated_Name:3] 6RZE_A.pdb	NGFLLDGFPR	TIPQADAMKE	AGINVDYVLE	FDVPDELIVD	
[Truncated_Name:4] 3HPR_A.pdb	NGFLLDGFPR	TIPQADAMKE	AGINVDYVLE	FDVPDELIVD	
[Truncated_Name:5] 1E4V_A.pdb	NGFLLDGFPR	TIPQADAMKE	AGINVDYVLE	FDVPDELIVD	
[Truncated_Name:6] 5EJE_A.pdb	NGFLLDGFPR	TIPQADAMKE	AGINVDYVLE	FDVPDELIVD	
[Truncated_Name:7] 1E4Y_A.pdb	NGFLLDGFPR	TIPQADAMKE	AGINVDYVLE	FDVPDELIVD	
[Truncated_Name:8] 3X2S_A.pdb	NGFLLDGFPR	TIPQADAMKE	AGINVDYVLE	FDVPDELIVD	
[Truncated_Name:9] 6HAM_A.pdb	NGFLLDGFPR	TIPQADAMKE	AGINVDYVLE	FDVPDELIVD	
[Truncated_Name:10] 8PVW_A.pdb	NGFLLDGFPR	TIPQADAMKE	AGINVDYVLE	FDVPDELIVD	
[Truncated_Name:11] 4K46_A.pdb	KGFLLDGFPR	TIPQADGLKE	VGVVVDYVIE	FDVADSVIVE	
[Truncated_Name:12] 4NP6_A.pdb	KGFLLDGFPR	TIPQADGLKE	MGINVDYVIE	FDVADSVIVE	
	*****~^**	*~	****~****	* ~**~	
	81	.	.	.	120
	121	.	.	.	160
[Truncated_Name:1] 1AKE_A.pdb	RIVGRRVHAP	SGRVYHVKFN	PPKVEGKDD	VDTGEELTTR	KD
[Truncated_Name:2] 6S36_A.pdb	KIVGRRVHAP	SGRVYHVKFN	PPKVEGKDD	VDTGEELTTR	KD
[Truncated_Name:3] 6RZE_A.pdb	AIVGRRVHAP	SGRVYHVKFN	PPKVEGKDD	VDTGEELTTR	KD
[Truncated_Name:4] 3HPR_A.pdb	RIVGRRVHAP	SGRVYHVKFN	PPKVEGKDD	GTGEELTTR	KD
[Truncated_Name:5] 1E4V_A.pdb	RIVGRRVHAP	SGRVYHVKFN	PPKVEGKDD	VDTGEELTTR	KD
[Truncated_Name:6] 5EJE_A.pdb	RIVGRRVHAP	SGRVYHVKFN	PPKVEGKDD	VDTGEELTTR	KD
[Truncated_Name:7] 1E4Y_A.pdb	RIVGRRVHAP	SGRVYHVKFN	PPKVEGKDD	VDTGEELTTR	KD
[Truncated_Name:8] 3X2S_A.pdb	RIVGRRVHAP	SGRVYHVKFN	PPKVEGKDD	VDTGEELTTR	KD
[Truncated_Name:9] 6HAM_A.pdb	RIVGRRVHAP	SGRVYHVKFN	PPKVEGKDD	VDTGEELTTR	KD
[Truncated_Name:10] 8PVW_A.pdb	RILKRGE--	TSGRV-----			D
[Truncated_Name:11] 4K46_A.pdb	RMAGRRAHLA	SGRTYHN	VYNPPKVEG	KDDVTGEDLV	IRED
[Truncated_Name:12] 4NP6_A.pdb	RMAGRRAHLA	PSGRTYH	VYNPPKVEG	KDDVTGEDLV	IRED
	^	*	***		*
	121	.	.	.	160
	161	.	.	.	200
[Truncated_Name:1] 1AKE_A.pdb	DQEETVRKRL	VEYHQMTAP	LIGYYSKEAE	AGNTKYAKVD	G
[Truncated_Name:2] 6S36_A.pdb	DQEETVRKRL	VEYHQMTAP	LIGYYSKEAE	AGNTKYAKVD	G
[Truncated_Name:3] 6RZE_A.pdb	DQEETVRKRL	VEYHQMTAP	LIGYYSKEAE	AGNTKYAKVD	G
[Truncated_Name:4] 3HPR_A.pdb	DQEETVRKRL	VEYHQMTAP	LIGYYSKEAE	AGNTKYAKVD	G
[Truncated_Name:5] 1E4V_A.pdb	DQEETVRKRL	VEYHQMTAP	LIGYYSKEAE	AGNTKYAKVD	G
[Truncated_Name:6] 5EJE_A.pdb	DQEECVRKRL	VEYHQMTAP	LIGYYSKEAE	AGNTKYAKVD	G
[Truncated_Name:7] 1E4Y_A.pdb	DQEETVRKRL	VEYHQMTAP	LIGYYSKEAE	AGNTKYAKVD	G
[Truncated_Name:8] 3X2S_A.pdb	DQEETVRKRL	CEYHQMTAP	LIGYYSKEAE	AGNTKYAKVD	G
[Truncated_Name:9] 6HAM_A.pdb	DQEETVRKRL	VEYHQMTAP	LIGYYSKEAE	AGNTKYAKVD	G
[Truncated_Name:10] 8PVW_A.pdb	DNEETVRKRL	VEYHQMTAP	LIGYYSKEAE	AGNTKYAKVD	G

```

[Truncated_Name:11]4K46_A.pdb  DKEETVLARLGVYHNQTAPLIAYYGKEAEAGNTQYLKFDG
[Truncated_Name:12]4NP6_A.pdb  DKEETVRARLNVYHTQTAPLIEYYGKEAAAGKTQYLKFDG
                                * ** *  **  **  ***** ** *** ** * * * **
                                161          .          .          .          200

                                201          .          216
[Truncated_Name:1]1AKE_A.pdb    TKPVAEVRADLEKILG
[Truncated_Name:2]6S36_A.pdb    TKPVAEVRADLEKILG
[Truncated_Name:3]6RZE_A.pdb    TKPVAEVRADLEKILG
[Truncated_Name:4]3HPR_A.pdb    TKPVAEVRADLEKILG
[Truncated_Name:5]1E4V_A.pdb    TKPVAEVRADLEKILG
[Truncated_Name:6]5EJE_A.pdb    TKPVAEVRADLEKILG
[Truncated_Name:7]1E4Y_A.pdb    TKPVAEVRADLEKILG
[Truncated_Name:8]3X2S_A.pdb    TKPVAEVRADLEKILG
[Truncated_Name:9]6HAM_A.pdb    TKPVCEVRADLEKILG
[Truncated_Name:10]8PVW_A.pdb    TKPVAEVRADLEKILG
[Truncated_Name:11]4K46_A.pdb    TKAVAEVSAELEKALA
[Truncated_Name:12]4NP6_A.pdb    TKQVSEVSADIAKALA
                                ** * ** *~* *~
                                201          .          216

```

Call:

```
pdbaln(files = files, fit = TRUE, exefile = "msa")
```

Class:

```
pdb, fasta
```

Alignment dimensions:

```
12 sequence rows; 216 position columns (187 non-gap, 29 gap)
```

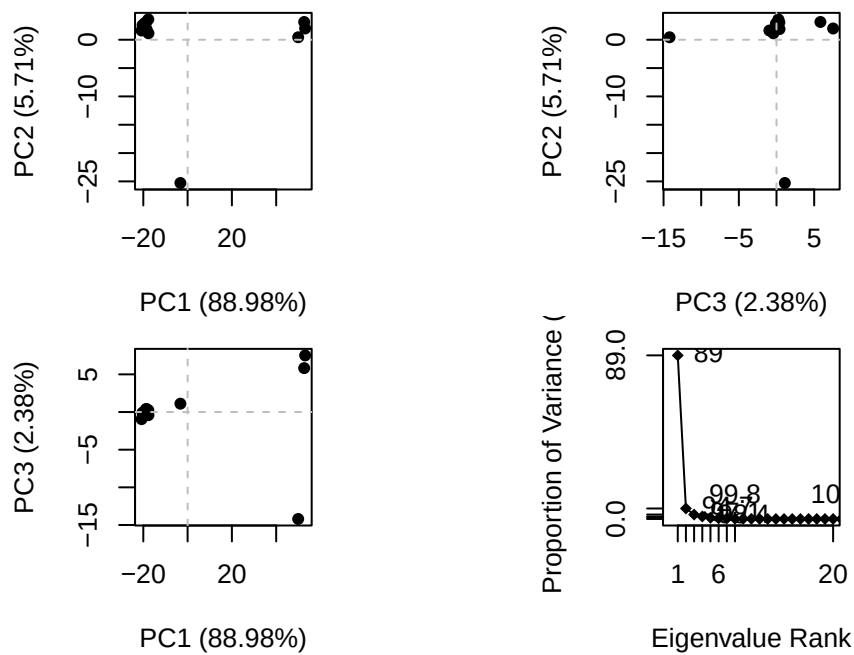
```
+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

Quick interactive structural view

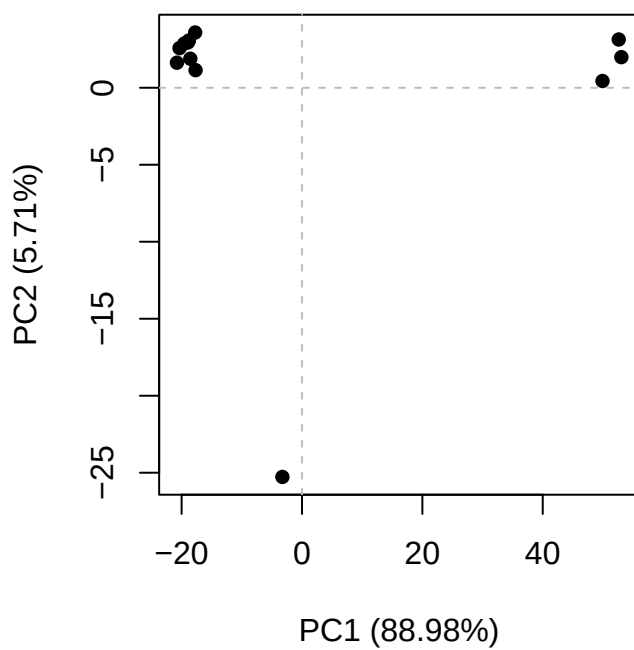
```
#view.pdb(pdb)
```

PCA of all this structural data (x, y, and z atom coordinates)

```
pc <- pca(pdb)
plot(pc)
```



```
plot(pc, 1:2)
```



Interactive view of the PC1 captured structural differences:

```
#view.pca(pc)
```

```
mktrj(pc, file="pca.pdb")
```

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